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REGULAR PAPER

LLMs applied to web scraping and web crawling: a systematic review

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Abstract

The integration of Large Language Models (LLMs) with web scraping and crawling techniques is transforming automated web data extraction by enabling semantic understanding and adaptability. This Systematic Literature Review (SLR) synthesizes evidence regarding this integration, focusing on tools, models, challenges, evaluation methods, trends, and applications. Following PRISMA guidelines, we conducted a rigorous search across Scopus, Web of Science, ACM, and IEEE databases (2021–2025). From 976 screened records, 91 high-quality studies (53 conference papers and 38 journal articles) were selected after duplicate removal, screening, and AI-powered quality assessment. The field has experienced explosive growth, with 84% of publications appearing in 2024–2025 alone (36 in 2024, 40 in 2025). Key tools include Scrapy, BeautifulSoup, and Selenium, with emerging LLM-augmented tools like Scrapeghost, Crawl4AI, and ScrapeGraphAI. While transformer-based models dominate (86 of 91 papers), the landscape is diversifying: the BERT family appears in 23 studies, the GPT family in 34, and other LLMs (Llama, Mistral, Claude, Gemini) in 44. Major challenges involve HTML complexity, computational costs, token limits, data biases, and legal risks. Evaluation relies on hybrid frameworks combining task-specific metrics (F1, BLEU, RAGAS), human validation, and operational efficiency measures. Applications span Cybersecurity, Healthcare, Education, E-commerce, Media, Technology, and Finance/Legal, with high thematic specialization. A notable trend is the shift toward efficient Small Language Models (SLMs) for resource-constrained, domain-specific tasks. The findings suggest that LLMs are enabling a decisive transition from rule-based to semantic, agentic approaches in web extraction. Challenges in robustness and efficiency persist, but trends point toward intelligent, domain-specialized, and ethically aware systems. Future work should explore SLM implementation, hybrid pipelines, and standardized evaluation benchmarks.

Extended author information available on the last page of the article

Keywords Web scraping · Web crawling · Large language models · Systematic review · Structured output

1 Introduction

The process of Information Extraction (IE) is used to extract useful information from unstructured or semi-structured data. The main goal of IE in textual data is to identify and annotate semantic information within the text and convert it into a structured format. The retrieved data can then be utilized in a data analysis pipeline to aid in data preparation before preprocessing and analysis [3].

While reliable Information Extraction (IE) techniques, such as those leveraging Machine and Deep Learning [5, 12, 111], are crucial for transforming unstructured data into a structured format for analysis, this research specifically addresses the foundational data acquisition phase. The primary focus of this study is to systematically review the methodologies, challenges, and tools associated with web scraping and web crawling. These processes are responsible for gathering the raw, unstructured data from the web that serves as the essential input for advanced IE systems. Therefore, a detailed examination of IE techniques themselves falls outside the scope of this paper.

Web scraping and web crawling are the automated process of extracting data from websites using specialized software [55, 71, 101]. Web scraping tools convert unstructured web data into a structured format, utilizing it for the purpose for which it is planned. Web crawling tools navigate through multiple pages to collect data [47]. This technique plays a crucial role in areas like Business Intelligence in today's digital era [98]. It enables the retrieval of structured information from textual content, including HTML. Web scraping is particularly valuable when data is not readily available in machine-readable formats such as JSON or XML [48]. This comparison can be seen in Fig. 1.

Despite their widespread adoption, traditional web scraping and crawling methodologies exhibit significant limitations that undermine their robustness and scalability. Conventional tools such as Scrapy, BeautifulSoup, and wget often rely heavily on static HTML parsing and pre-defined DOM selectors, rendering them highly susceptible to structural changes in website layouts [100]. Furthermore, these approaches fundamentally lack semantic reasoning capabilities, preventing them from understanding contextual nuances or adapting to dynamically generated content without manual intervention. This inherent fragility and inability to process web content at a conceptual level severely constrains their effectiveness in modern, JavaScript-heavy web environments, thereby creating an imperative need for more intelligent and adaptive solutions.

Large Language Models (LLMs) exhibit intrinsic properties that position them as optimal tools for web scraping and crawling tasks. Their transformer-based architecture enables efficient processing of extensive text sequences and complex HTML structures, extracting semantic and contextual patterns with high fidelity. While LLMs excel in general-purpose understanding, Small Language Models (SLMs), such as BERT and RoBERTa, offer a complementary advantage through parameter effi-

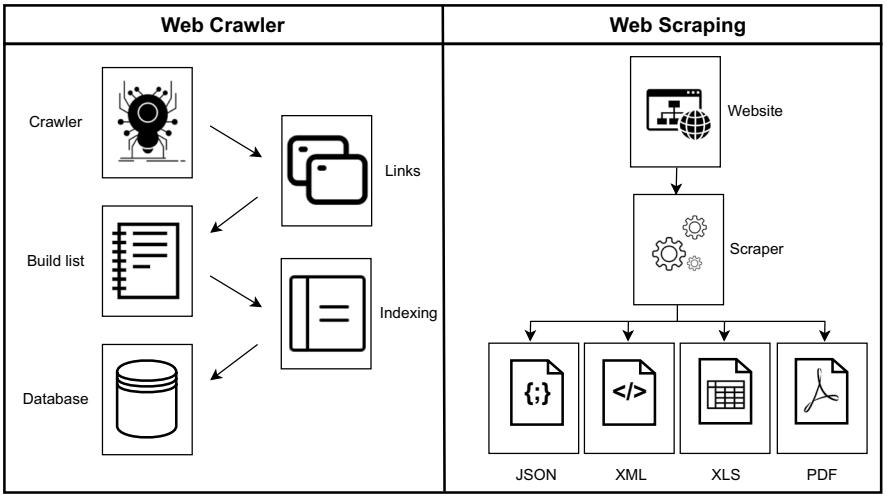


Fig. 1 Comparison between web crawler and web scraping. Based on [113]

ciency and cost-effective fine-tuning for task-specific extraction [65]. For example, in recommendation systems, LLMs generate item descriptions from titles through in-context learning, replicating the style of traditionally scraped data with high metric alignment [1]. This capability reduces the reliance on manual extraction techniques, mitigating data inconsistencies and biases inherent in conventional scraping.

This study conducts a Systematic Literature Review (SLR) focused on research addressing web scraping and web crawling techniques from 2021 to 2025. This period captures the rise and industrial adoption of Transformer-based architectures (the foundation of LLMs) marking a shift from rule-based and classical Machine Learning approaches to semantically grounded, autonomous data acquisition methods. The objective of this SLR is to identify and categorize the dominant LLM architectures that are redefining web data acquisition, contrasting their adoption, performance, and efficiency across different implementations. Limiting the review to this timeframe ensures methodological coherence and reflects the conceptual transition toward LLM-driven reasoning and navigation in web data extraction.

This SLR represents, to the best of our knowledge, the first systematic effort that rigorously delimits its scope to the data acquisition stage, specifically web scraping and web crawling, rather than addressing Information Extraction as a whole. Prior reviews and surveys often treat scraping as a preliminary or auxiliary component within broader IE pipelines. In contrast, this study isolates and examines scraping and crawling as independent and technically complex stages that define the quality, structure, and reliability of subsequent data-driven analysis.

The remainder of the paper is presented as follows: Sect. 2 describes the materials and methods used in the literature review. Section 3 presents the results obtained and the answers to the research questions. Section 4 discusses and presents the synergy with specific case studies. Section 5 describes the limitations of this study. Finally, conclusions and future research directions are presented in Sect. 6.

2 Materials and methods

This SLR follows the guidelines provided by Petersen et al. [75] and Kitchenham [50], in addition to the criteria suggested by the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) statements [73, 74].

The SLR conducted focused on the use of LLM to extract information from websites using web scraping and web crawling techniques, which is essential for various tasks involving information gathering and the subsequent use of this extracted information. Therefore, it aims to answer the following research questions:

- RQ_1: What are the tools that have been used for web scraping and web crawling?
- RQ_2: Which LLMs have been used in studies on web scraping and web crawling in the last five years?
- RQ_3: What are the main challenges and limitations when integrating LLMs into web scraping and web crawling workflows?
- RQ_4: How do you evaluate the effectiveness and quality of data extraction when using LLMs in web scraping and web crawling?
- RQ_5: What patterns or trends are observed in the evolution of LLMs applications for web scraping/crawling over time (beyond the number of publications)?
- RQ_6: In which specific domains or industries are LLMs for web scraping and web crawling being most actively applied?
- RQ_7: What characteristics of Small Language Models (SLMs) that could be special purpose models make them particularly suitable or challenging for web scraping and web crawling tasks, and how are they currently being used or expected to be used?

In order to achieve the proposed objective and answer the research questions, a search was carried out in the main Computer Science databases such as: Scopus, Web of Science, ACM and IEEE. In order to filter the search, some inclusion and exclusion criteria have been established and are detailed in Table 1.

Table 1 shows that it has been decided to consider only papers published in English-language, journals and conference papers since 2021. This decision has been made in order to compile the most relevant papers, although some potentially important papers published as preprints may be excluded.

Based on the PRISMA flow chart presented in Fig. 2, the three phases of identification, screening and inclusion were applied. Similar search strings were used for the databases mentioned before. Table 2 shows the search strings used and the number of papers found for each of them.

In the identification phase, 1170 papers were found, distributed as follows: in Scopus 768, in Web of Science 101, in ACM 30 and in IEEE 271. All these records were

Table 1 Inclusion and exclusion criteria

Criteria	Inclusion	Exclusion
Publication year	2021 - 2025	Before 2021
Document type	Journal papers, Conference papers	Thesis and preprints
Language	English	Not English

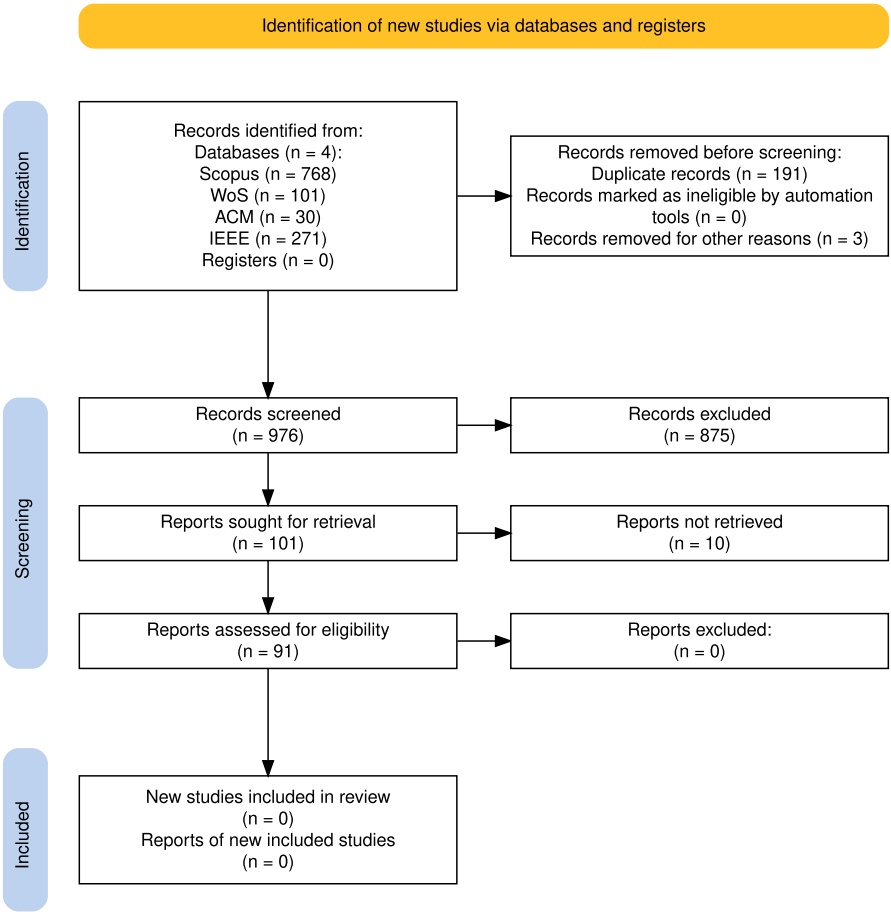


Fig. 2 PRISMA flow chart. Elaborated with tool PRISMA Flow Diagram [34]

exported to a spreadsheet for analysis, and 191 duplicate papers were removed, leaving a total of 976 papers for the Screening phase. The table with complete information on the primary studies found is available in Online Resource 1 [56]. In this second phase, an AI-powered quality assessment was conducted using advanced language models such as Qwen 2.5 Plus, DeepSeek-R1, and GPT-4o. Ten evaluation questions, adapted from the framework proposed by Kitchenham et al. [51], were defined to serve as the quality assessment criteria. Each model independently evaluated the selected studies based on these questions, generating preliminary scores and qualitative summaries. The prompt used by each model to evaluate the papers is available in Online Resource 1 [56]. The scores obtained were collected in a spreadsheet for further processing. Subsequently, a human validation stage was performed, in which the agreement between the AI-generated and human-assessed evaluations was analyzed. To resolve disagreements, some papers were reviewed in detail by researchers to corroborate the results of the scores obtained by the LLMs or, failing that, to discard the paper for the final dataset. This hybrid approach demonstrates that the AI-assisted

Table 2 Search string results

Database	String	Papers
Scopus	(TITLE-ABS-KEY ("large language model*" OR "LLM*" OR "foundation model*" OR "language model*") AND ALL ("web scrap*" OR "web crawl*")) AND PUBYEAR > 2020 AND PUBYEAR < 2026 AND (LIMIT-TO (DOCTYPE, "cp") OR LIMIT-TO (DOCTYPE, "ar")) AND (LIMIT-TO (LANGUAGE, "English"))	768
WoS	TS=(("large language model*" OR "LLM*" OR "foundation model*" OR "language model*") AND ("web scrap*" OR "web crawl*")) AND PY=(2021-2025) AND DT=(Article OR "Proceedings Paper") AND LA=(English)	101
ACM	AllField:([[All: "large language model*"] OR [All: "LLM*"] OR [All: "foundation model*"] OR [All: "language model*"]] AND [[All: "web scrap*"] OR [All: "web crawl*"]] AND [Publication Date: (01/01/2021 TO 12/31/2025)] AND [Publication Type: (Journals OR Proceedings)] AND [Content Type: Research Article])	30
IEEE	("All Metadata":"large language model*" OR "All Metadata":"LLM*" OR "All Metadata":"foundation model*" OR "All Metadata":"language model*") AND ("All Metadata":"web scrap*" OR "All Metadata":"web crawl*")	271

Table 3 Inter-rater agreement metrics between AI models and human validation

AI Model	Agreement Rate (%)	Cohen's Kappa (κ)	Interpretation*
Qwen 2.5 Plus	96%	0.77	Substantial agreement
DeepSeek-R1	93%	0.67	Substantial agreement
GPT-4o	94%	0.71	Substantial agreement
Average	**94%**	**0.72**	**Substantial agreement**

*Kappa interpretation follows Landis and Koch criteria [57]: $<$ 0.20 = slight; 0.21-0.40 = fair; 0.41-0.60 = moderate; 0.61-0.80 = substantial; 0.81-1.00 = almost perfect

quality assessment acts as an accelerator of the review process by enhancing consistency and efficiency, rather than as a replacement for human judgment.

To quantify the consistency between automated AI assessments and human expert validation, we calculated two inter-rater agreement metrics: simple agreement rate and Cohen's kappa coefficient. The agreement rate was computed as the percentage of studies where the binary inclusion/exclusion decision (threshold $\geq$7) aligned between each AI model and the human evaluator. Cohen's kappa coefficient provides a more rigorous measure of agreement adjusted for chance expectancy [57]. As shown in Table 3, the results demonstrate substantial agreement between all AI models and human validation, with agreement rates ranging from 93% to 96% and kappa values between 0.67 and 0.77. This validates the reliability of the AI-assisted

assessment approach as a consistent accelerator of the review process, with all models showing substantial alignment with human judgment.

Table 4 describes the questions and the corresponding evaluation criteria, i.e., each question is evaluated with 1 point if the question is answered satisfactorily, 0.5 if it is partially answered and 0 if it is not answered at all.

Only studies scoring 7 or higher in the AI-assisted evaluations were included, following strict criteria emphasizing empirical evidence, primary LLM application, and measurable outcomes. Of the 101 initially identified papers, ten were unavailable, resulting in 91 final inclusions. This low inclusion rate reflects the rigor of the selection process and reinforces the quality of the corpus, which represents the most robust and relevant research within this specialized domain. Table S 1 presents these papers, and it is given in Online Resource 2 [56].

3 Results

This SLR study of papers published in journals between 2021 and 2025 ensures the inclusion of recent studies using the latest technology in web scraping and web crawling techniques using LLMs.

The analysis of the **annual distribution** of identified and analyzed publications, as illustrated in Fig. 3, reveals a pronounced and accelerating interest in the application of large language models (LLMs) to web scraping and crawling tasks over the five-year period from 2021 to 2025.

The data indicates a modest but non-negligible level of research activity in the early years: 3 publications in 2021 (3.3% of the total corpus), followed by 6 in 2022 (6.6%). This initial phase likely reflects exploratory efforts by researchers to assess

Table 4 Qualitative checklist

No.	Question
Q1	Are the objectives of the study clearly related to web scraping or web crawling using LLMs?
Q2	Does the methodology thoroughly describe the web scraping/crawling process and LLM integration?
Q3	Does the study clearly specify the research type (e.g., case study, experiment, applied research)?
Q4	Do the conclusions specifically address the results and effectiveness of web scraping/crawling with LLMs?
Q5	Does the study specify the structuring format of the extracted data (json, markdown, yaml, etc.)?
Q6	Does the study specify which LLM model(s) and version were used?
Q7	Are the specific tools and libraries used for web scraping/crawling described?
Q8	Does the study clearly describe the practical use case or application of the web scraping/crawling?
Q9	Does the study analyze the advantages and limitations of using LLMs in web scraping/crawling?
Q10	Does the study detail the URL input method and management process?

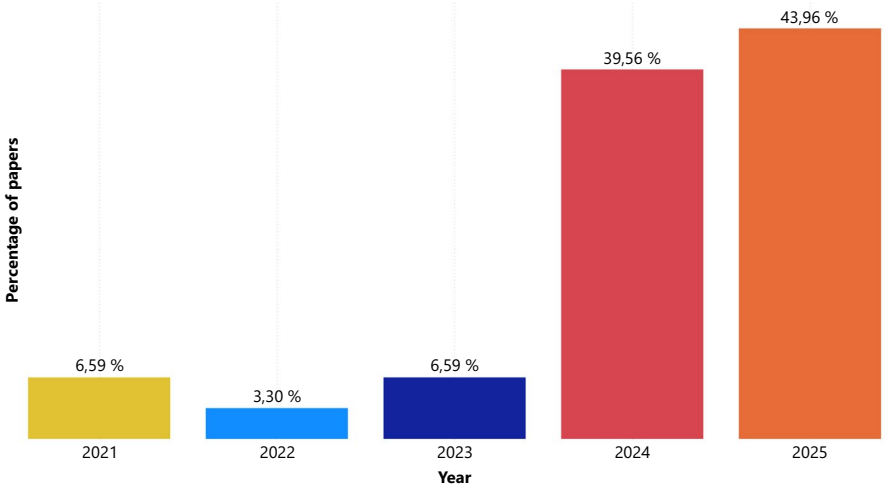


Fig. 3 Percentage of papers per year

the theoretical potential of emerging LLMs for addressing persistent challenges in web automation–such as parsing heterogeneous website structures, interpreting unstructured or semi-structured HTML content, and adapting to dynamic page layouts without brittle rule-based extractors.

Research output remained stable in 2023, with another 6 publications (6.6%), suggesting a period of methodological refinement and tool development that laid the groundwork for broader adoption. The real inflection point occurs in 2024, which accounts for 36 papers (39.6% of the dataset), followed by an even more substantial contribution in 2025 with 40 papers (44.0%). Together, these two years represent over 83% of all identified literature on the topic.

This dramatic surge is not an anomaly but the direct consequence of the widespread availability, commercialization, and maturation of powerful generative AI platforms and their APIs–such as OpenAI's GPT-4, Anthropic's Claude, and Meta's LLaMA 2–throughout late 2023 and into 2025 [17]. The research community has shifted decisively from conceptual feasibility studies to practical implementations, empirical evaluations, and system-level integrations. The volume and velocity of publications in 2024 and 2025 confirm that LLM-enhanced web extraction has transitioned from a niche idea to a mainstream paradigm in computer science, offering unprecedented levels of robustness, adaptability, and semantic awareness in automated data harvesting systems.

Figure 4 shows the distribution of **data collection techniques** addressed in the reviewed literature: web scraping, web crawling, or a combination of both. The majority of studies focus exclusively on web scraping (55 papers, 60.44%), followed by those dedicated solely to web crawling (29 papers, 31.87%). Only 7 papers (7.69%) integrate both techniques. This distribution reveals a strong tendency toward thematic specialization: 92.31% of the works address only one technique, while fewer than 8% explore their combined use–suggesting that research in this domain remains largely siloed rather than integrative.

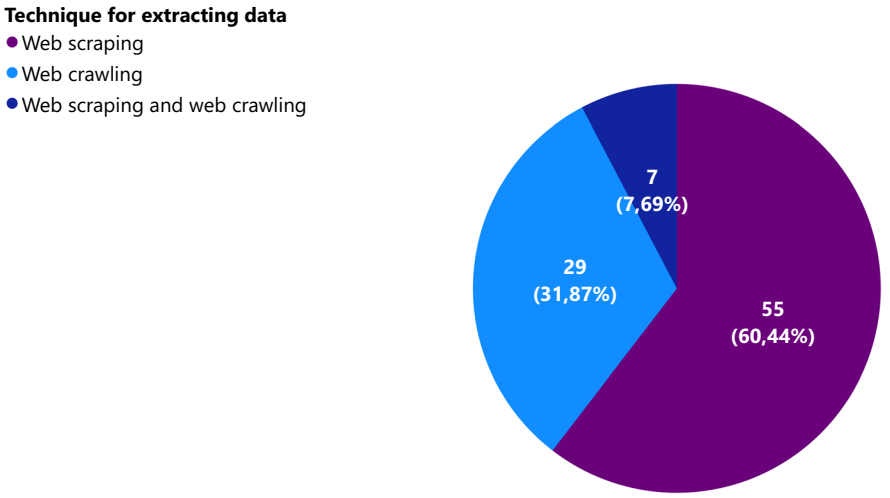


Fig. 4 Techniques found

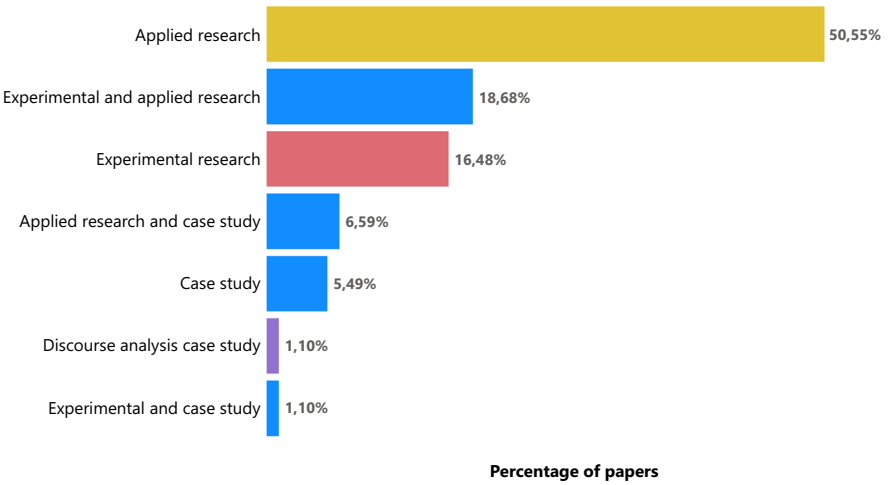


Fig. 5 Types of research papers found

Figure 5 summarizes the **methodological approaches** across the 91 reviewed papers. Applied research dominates the landscape, appearing in 75.8% of studies–either as standalone work (50.55%), combined with experimental methods (18.68%), or integrated into case studies (6.59%). Pure experimental research accounts for 16.48%, while other designs (e.g., discourse analysis) are rare. This prevalence reflects the practical orientation of LLM-based web scraping and crawling, which is often driven by industrial needs such as efficiency, scalability, and legal compliance. The subset combining applied and experimental methods (18.68%) seeks to balance real-world utility with scientific validation–demonstrating not only that a solution

works in practice, but also why and how it outperforms alternatives under controlled conditions.

Figure 6 shows the distribution of **output data formats** reported across the 91 reviewed papers. JSON is the most commonly used format (38.46%), favored for its lightweight structure, human readability, and seamless integration with modern APIs and NoSQL databases. CSV follows at 8.79%, typically chosen for its simplicity and compatibility with spreadsheet software and basic analytics workflows. All other formats (XML, WET files, RDF, BSON, TFRecord, and Markdown) each appear in only 1.10% of studies, indicating niche or domain-specific applications (e.g., XML for legacy interoperability, WET for web archiving, BSON for high-volume storage). Notably, 41.76% of papers do not specify an output format, which may reflect either omissions in methodological reporting or that format choice was peripheral to the study's goals.

To answer $\mathbf{RQ}_1$, Table 5 shows the **tools used for web scraping and crawling**. Traditional libraries like BeautifulSoup, Requests, Scrapy, and Selenium, remain dominant due to their reliability and flexibility. Specialized crawlers like ACHE and datasets like Common Crawl support focused or large-scale tasks. Importantly, a growing number of works adopt LLM-powered tools (Scrapeghost, ScrapeGraphAI, Crawl4AI) that generate scrapers from natural language prompts. This signals a shift toward smarter, low-code extraction systems that adapt to site changes with minimal manual intervention–marking an evolution from rigid, selector-based approaches to more resilient, AI-driven pipelines.

To answer $\mathbf{RQ}_2$, Table S2 summarizes the LLMs used across 86 of the 91 reviewed papers (see Online Resource 2 [56]). The landscape is dominated by transformer-based models, grouped into three families:

- The **BERT family** (BERT, RoBERTa, DistilBERT, SBERT, and domain-specific variants like BatteryBERT or CamemBERT) appears in 23 papers, primarily for

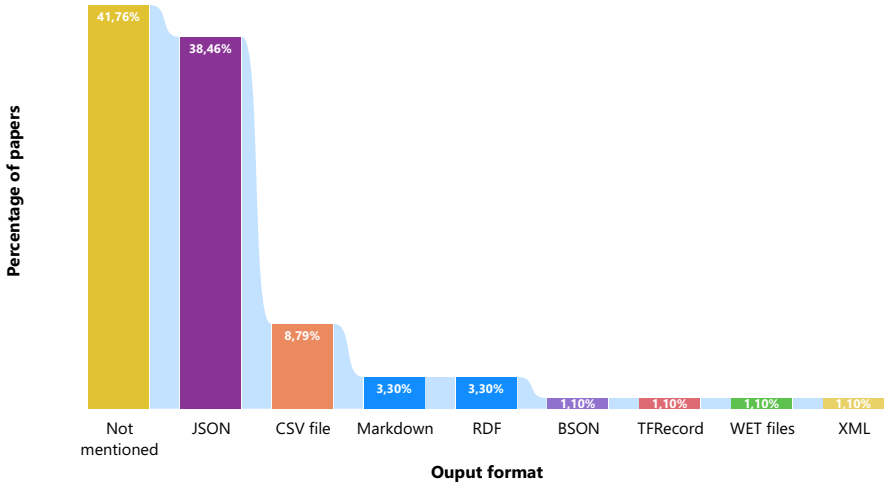


Fig. 6 Data extraction formats that are mentioned in the papers

Table 5 Tools used for web scraping or web crawling in journal and conference publications

Tool	Use and References
Common Crawl	Pre-existing large-scale web crawl dataset [20, 21, 33]
BeautifulSoup	HTML/XML parser for structured data extraction [2, 8, 10, 14, 15, 23, 24, 28, 29, 32, 38, 40–42, 62, 63, 66, 69, 72, 78, 82, 84, 87, 90, 94, 105, 110, 114]
Scrapy	Framework for scalable crawling and scraping [2, 6, 13, 15, 19, 24, 26, 27, 31, 40, 42, 49, 67, 69, 70, 72, 80, 83–85, 87, 93, 97, 102, 103, 108, 112]
Selenium	Browser automation for dynamic/JavaScript-heavy sites [2, 10, 19, 27, 28, 40, 42, 44, 49, 69, 70, 72, 83, 88, 89, 91, 93, 96, 107, 109, 110]
Requests	HTTP library for fetching static web content [2, 10, 19, 27, 28, 42, 44, 58, 66, 69, 72, 90, 91, 93, 94, 110]
ACHE crawler / SeedFinder	Focused crawler with seed URL selection [53]
Bitextor	Parallel corpus extraction from multilingual websites [60, 63]
WebHarvy	Visual, GUI-based scraping tool [25]
PyMuPDF	Text extraction from PDF documents (post-scraping) [77]
Akka	Concurrency framework for distributed crawling [86]
Chrome DevTools Protocol	Low-level browser control for advanced crawling [52]
Scrapeghost	LLM-powered code generation for scraping tasks [79]
Crawljax	Crawler for dynamic web applications and form testing [13]
Puppeteer	Headless browser automation (Node.js) [107]
DOMDocument / DOMXPath	Native PHP tools for DOM parsing [9]
rvest	R package for web scraping [43]
Trafilatura	Robust text extraction from HTML/news pages [54, 105]
Playwright	Modern browser automation alternative to Selenium [10, 19, 40, 83]
StormCrawler	Scalable, Apache-based web crawler [21]
markdownify	HTML-to-Markdown conversion for clean text extraction [89]
WebBaseLoader (LangChain)	Generic web page loader for RAG pipelines [49]

semantic understanding, classification, and information extraction from structured or semi-structured web content.

- The **GPT family** (GPT−3.5, GPT-4, GPT-4o, and their variants) is used in 34 papers, often to automate scraping logic, generate code, interpret dynamic interfaces, or enable agentic behavior through prompt engineering.

- **Other LLMs**: Llama, Mistral, Claude, Gemini, and DeepSeek are employed in 44 papers, reflecting a rapid diversification toward open-weight and multimodal models that support customization, cost efficiency, and advanced reasoning.

Five papers were excluded as they rely on non-LLM NLP techniques: FastText and Word2Vec for static embeddings [33, 53, 86], traditional n-gram models [59], or the Universal Sentence Encoder [60], which lack generative or contextual adaptation capabilities.

To answer $\mathbf{RQ}_3$, our analysis of 91 studies reveals that integrating LLMs into web scraping and crawling introduces persistent challenges across four interrelated dimensions: technical robustness, data quality and bias, computational and economic feasibility, and ethical-legal constraints.

- **Technical robustness remains limited**. LLMs struggle with the structural complexity of real-world web content–dynamic JavaScript, inconsistent HTML, CAPTCHAs, and adversarial obfuscation (e.g., hidden links via CSS or transparent images) frequently break extraction pipelines [10, 102, 104]. Critically, LLMs lack inherent visual grounding and DOM reasoning capabilities, making them prone to errors when generating XPaths or interpreting rendered layouts [39, 46]. This leads to fragile workflows that fail under minor UI changes [88, 114].
- **Data quality and systemic biases constrain reliability**. Web corpora exhibit domain, linguistic, and demographic imbalances–underrepresentation of low-resource languages [33], skewed source coverage [62], and noisy or unstructured inputs (e.g., PDFs, social media posts) demand extensive preprocessing [77, 86]. Moreover, LLMs often hallucinate plausible but incorrect facts–such as false locations [66], medical advice [30], or non-functional code [9]–especially when context is fragmented due to token limits [52, 94].
- **Computational and economic barriers hinder scalability**. High API costs for proprietary models (e.g., GPT-4, Claude) and GPU demands for local deployment make large-scale operations expensive [13, 61, 107]. Many teams resort to smaller or open-weight models (e.g., Llama-3, Mistral) to balance cost and performance [28, 93], yet this trade-off often sacrifices accuracy or latency–particularly in real-time or agent-based systems [39, 114].
- **Ethical and legal risks are pervasive**. Scraping violates Terms of Service on many platforms (e.g., LinkedIn, news sites) [27, 84], while using external LLM APIs risks data leakage in sensitive domains like OSINT or legal analysis [108]. Additionally, LLM content filters may block prompts for ethically critical tasks (e.g., phishing or dark web analysis) [52, 78], and copyright concerns persist when repurposing scraped content [4, 79].

In summary, LLM integration for scraping remains limited by fragility, cost, and compliance issues. This indicates that the field is still transitioning from experimental models to robust, production-ready systems.

To answer $\mathbf{RQ}_4$, evidence from 91 papers indicates that the effectiveness of LLM-enhanced web scraping and crawling is evaluated through a hybrid framework combining automated metrics, human validation, operational benchmarks, and task-

specific utility. his comprehensive approach guarantees that systems are not only accurate but also practical and efficient.

- **Automated quality metrics**: Most studies use standard NLP and information retrieval metrics, such as F1, precision, recall, and accuracy, to evaluate extracted data fields. [39, 46, 52, 94]. For structured extraction, Exact Match (EM) and ROUGE-L assess field-level fidelity [10, 78], while semantic similarity (e.g., BERTScore) evaluates contextual correctness beyond lexical overlap [36, 87]. In retrieval-augmented pipelines, frameworks like RAGAS measure faithfulness and context relevance [72, 87].
- **Human and expert validation** is widely used to ground results in reality. This includes manual annotation of ground-truth datasets [14, 63], expert ratings on Likert scales [30], and domain-specific audits (e.g., legal or medical fact-checking) [15, 32]. User studies further assess perceived helpfulness and decision-making impact [28, 89].
- **Operational and efficiency metrics** quantify practical viability. These include execution time, latency, throughput (e.g., pages/minute) [9, 89], cost per query (USD/EUR) [36, 79], and scalability (e.g., number of records processed or HTTP success rates) [23, 108].
- **Task and domain-specific validation** ensures real-world relevance. Examples include spatial correlation with population data [54], code coverage in security testing [107], alignment with official indices [43], or business KPIs like reduction in Mean Time to Respond (MTTR) [110]. Notably, many studies validate outputs against external signals (e.g., Google Trends, Bloomberg databases) rather than raw HTML, reflecting a shift toward utility-driven evaluation.

In summary, the field has moved beyond isolated accuracy metrics and now uses integrated, application-aware assessment. This approach balances statistical rigor, human judgment, system performance, and domain validity.. Table 6 summarizes these findings.

Table 6 Evaluation strategies for LLM-based web scraping and crawling

Dimension	Key metrics & methods	Representative papers
Automated quality	F1, P/R, Accuracy; EM, ROUGE-L; BERTScore; RAGAS (faithfulness, relevance)	[10, 39, 46, 52, 78, 87, 94]
Human validation	Manual annotation; Expert ratings (Likert); Domain audits; User studies	[14, 15, 28, 30, 32, 63, 89]
Operational efficiency	Latency, throughput, cost (USD); HTTP success rate; Scalability (records processed)	[9, 23, 36, 79, 89, 108]
Task-specific utility	Spatial correlation (R^2); Code coverage; MTTR reduction; Alignment with official indices or external benchmarks (Google Trends, Bloomberg)	[43, 45, 54, 90, 107, 110]

To address **RQ**$_5$, the review of 91 studies shows a clear evolutionary path in web scraping: systems are transitioning from rigid, rule-based approaches to adaptive, semantic, and ethical architectures. This transformation is qualitative in nature, defined by five key, interconnected trends.

- **From passive extraction to agentic autonomy**. Early systems relied on static selectors or regex; today's LLMs act as autonomous agents that generate executable code (e.g., XPaths, form-fillers), navigate dynamic sites, and self-repair when layouts change [13, 39, 46, 107, 114]. This shift allows interaction with complex interfaces, such as CAPTCHAs, multi-step forms, and dark web marketplaces, which were previously unreachable for traditional scrapers.
- **Deep integration into end-to-end pipelines**. LLMs are no longer isolated post-processing tools but core components spanning data acquisition, cleaning, augmentation, and reasoning. Systems now embed RAG, vector databases, and multimodal inputs (text and screenshots) to ground extraction in context [2, 19, 28, 102, 108].
- **Efficiency-driven pragmatism**. As costs and latency become critical, researchers favor smaller, open-weight models (Llama-3, Mistral, Gemma) over proprietary APIs, optimize prompts for token efficiency, and deploy hybrid architectures that combine LLMs with lightweight rules (regex, CSS selectors) to balance accuracy and speed [44, 61, 84, 89, 93].
- **Domain specialization and ethical grounding**. Generic scraping is being replaced by specialized workflows in fields like legal and clinical data. In these areas, LLMs are fine-tuned to ensure regulatory compliance, minimize bias, and respect data provenance [16, 30, 32, 54, 81].
- **Robustness through multimodality and hybrid intelligence**. To overcome HTML obfuscation and visual deception, modern systems fuse textual, structural, and visual signals (e.g., GPT-4V, Playwright screenshots) and validate outputs via human-in-the-loop or external benchmarks (Google Trends, official indices) [10, 45, 52, 66, 104].

Together, these trends signal a field maturing beyond proof-of-concept toward production-ready, responsible, and intelligent data extraction. Table S 3 synthesizes these patterns, and it is given in Online Resource 2 [56]

To answer **RQ**$_6$, LLM-enhanced web scraping and crawling are most actively applied in the following domains:

- **Cybersecurity & threat intelligence**: Dominant for threat detection [52, 112], dark web monitoring [78, 102], OSINT [83, 108], and vulnerability analysis [107]. High-stakes, data-rich environments drive adoption.
- **Healthcare & public health**: Applied to medical research [15, 25], clinical decision support [30], public health surveillance [63, 66], and legal/ethical compliance in medical data [32].
- **Education & research**: Includes educational chatbots [2, 18], academic knowledge graphs [19, 80], literature synthesis [15], and institutional Q&A systems [70, 89].

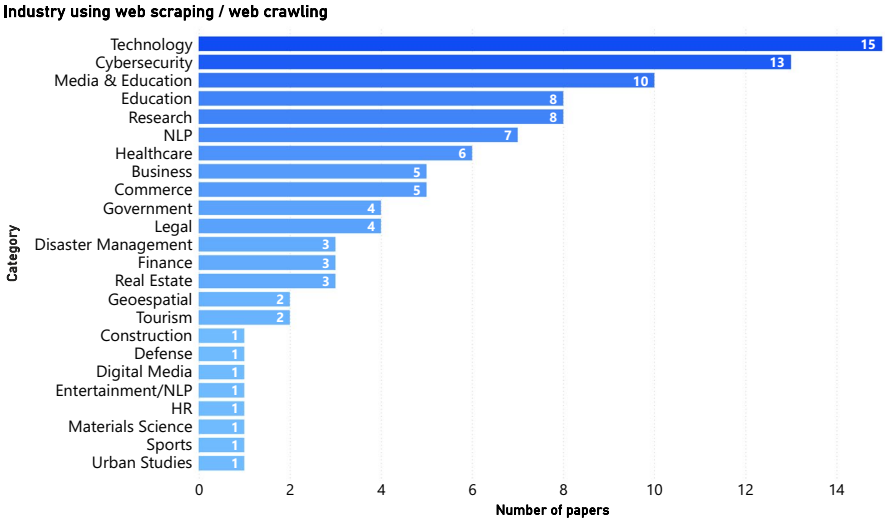


Fig. 7 Industry using web scraping / web crawling

- **E-commerce & business intelligence**: Used for product extraction [28, 36], brand analysis [84], market intelligence [90], and supply chain transparency [45].
- **Media, news & social analysis**: Covers sentiment analysis [44, 109], fake news detection [39, 62], and news aggregation [21, 103].
- **Technology & software engineering**: Focuses on automated testing [13, 107], code generation [79], web architecture [88], and infrastructure monitoring [61, 67].
- **Finance & legal**: Includes financial literacy tools [22], fraud detection [52], contract/privacy policy analysis [32, 39], and regulatory compliance [16, 77].

Beyond these core sectors, emerging applications span **Real Estate** [43, 69], **Disaster Management** [8, 66], **Geospatial Services** [24, 41], and **Human Rights** [29]. The field is highly fragmented across more than 25 sub-domains, most represented by only one or two papers. This highlight significant potential for methodological transfer and consolidation across these areas Fig. 7.

To answer $\mathbf{RQ}_7$, the analysis shows that SLMs, which typically have fewer than 10B parameters, are increasingly used for web scraping due to their efficiency, ease of deployment, and domain adaptability. Their use is characterized by the following trends:

- **Efficiency and sustainability**: SLMs like all-MiniLM, DistilBERT, and Mistral-7B enable faster inference, lower energy consumption, and reduced API costs, making them ideal for large-scale or real-time pipelines [23, 76, 82].
- **Domain specialization**: Fine-tuned SLMs (e.g., BatteryBERT for materials science, CamemBERT for French texts, Llama3-8B for geospatial or industrial data) outperform general LLMs in niche extraction tasks by reducing noise and improving accuracy [41, 62, 67, 95].

- **Edge and on-premise deployment**: SLMs support local execution, which is critical for privacy-sensitive domains such as legal or OSINT, as it eliminates cloud dependencies and data leakage risks [32, 96, 108].
- **Hybrid architectures**: SLMs are frequently used as initial routers for entity tagging or URL classification, whereas larger LLMs handle complex generation in order to optimize the balance between cost and performance [38, 89, 91].
- **Challenges**: Despite their benefits, SLMs lack contextual depth and multilingual robustness. This often requires aggressive preprocessing or fallback mechanisms due to their difficulty in handling dynamic content like CAPTCHAs or obfuscated HTML [10, 46, 52].

In summary, SLMs are strategic elements of sustainable scraping systems where factors like cost, latency, and data sovereignty are prioritized over larger model capabilities.

4 Discussion

This SLR exists within a body of literature that has previously surveyed the general field of web scraping and crawling. Foundational surveys, such as those by Khder [48] and Sharma [92], provide comprehensive overviews of classical techniques, tools, applications, and the enduring legal and ethical challenges. These works expertly catalog the landscape of traditional scraping technologies like BeautifulSoup, Scrapy, and Selenium, and their applications in domains like market research and competitive analysis [7, 48]. However, by nature of their publication date and scope, they primarily cover the pre-LLM era and thus cannot address the recent, paradigm-shifting integration of LLMs. Our study builds directly upon this foundational work but identifies and fills a critical gap: the lack of a systematic, empirical analysis of how LLMs are transforming web data extraction practices, the new challenges they introduce (token costs and hallucinations), and the emerging trends they enable.

Consequently, this review offers a significant update and specialization. While previous surveys discuss the potential of AI in broad terms, our findings (the surge in publications, the dominance of the BERT family and rise of GPT models, the hybrid evaluation frameworks) provide concrete, evidence-based insights into this rapid evolution. Furthermore, our methodological approach differentiates this work. Whereas prior reviews are descriptive and qualitative, we employ a rigorous SLR methodology guided by PRISMA criteria, utilizing AI-assisted quality assessment to objectively select and evaluate a focused corpus of literature. This allows us to move beyond a high-level overview to a quantified analysis of trends, tool prevalence, and research types, offering a more granular and evidence-driven perspective on the state of the art in LLM-powered web data acquisition.

Previous literature reviews on web scraping [7, 48, 92] have thoroughly documented the classical architectures, techniques, and applications that defined the field for years. These works establish a vital baseline, highlighting tools like Scrapy and challenges like CAPTCHAs and rate-limiting. Our systematic review complements and extends this existing body of knowledge by focusing exclusively on the disrup-

tive influence of LLMs. The results presented in this paper confirm that the core concepts and applications remain relevant; however, they are now being augmented and, in some cases, entirely redefined by LLMs.

In a study on advances in NLP with the use of Pre-trained Language Models [64], three paradigms are presented such as pre-training and fine-tuning models, prompting, and NLP as text generation. This article focuses on a wide range of NLP tasks that are not generative in nature (e.g., classification, sequence, and structure labeling). NLP tasks including parsing, information extraction (IE), question answering (QA), textual linking (TE), etc. answer (QA), textual explanation (TE), etc.

Another study [99] synthesizes techniques to optimize each stage of the NLP flow. The results include data efficiency strategies (filtering, active learning), model designs with reduced attention (sparse transformers, mixture of experts), fine-tuning methods with low parametric cost (adapters, prompt-tuning), and compression through pruning, quantization, and distillation. Although progress has been made in reducing costs without sacrificing performance, challenges remain, such as the management of long sequences, instability in sparse models and trade-offs between efficiency, fairness and robustness.

The paper of Yadav et al. [106] presents a comprehensive review of state-of-the-art approaches in extractive text summarization (ETS), classifying and analyzing methods according to their techniques, performance and limitations. Methods based on neural networks, graphs (PageRank, TextRank) and fuzzy logic are identified as the most effective, although challenges such as redundancy, consistency and adaptation to multiple languages remain. Evaluation measures (ROUGE, BLEU) and standard datasets (DUC, TAC) are discussed, noting the need for improved scalability and integration of advanced semantics for smoother and more accurate summaries.

This survey on web page classification by Hashemi [35] states that while the field is growing in importance due to the vast amount of online data, it is still in its early stages. Key challenges include ignoring metadata, underusing HTML structure, the high computational cost of image analysis, and a lack of exploration into semi-supervised learning. The study reviews current methods, points out these research gaps, and calls for better evaluation benchmarks.

An important study [37], mentioned how Deep Web crawling involves navigating dynamically generated pages that require submitting queries through search forms. Unlike traditional crawlers, these must detect and fill out forms to access hidden content. Existing research lacks a unified framework for comprehensive comparison of the latest crawling techniques. This study proposes such a framework, highlighting unexplored features and emphasizing the need for standardized evaluation methods.

The findings of this SLR are validated through alignment with contemporary case studies that exemplify RQ_3 (challenges in LLM integration) and RQ_6 (domain applications). Cavero et al. [11] demonstrate LLM-driven transformation of SaaS pricing structures, a domain identified in RQ_6 as of high impact for e-Commerce. Their work directly addresses SLR-documented RQ_3 challenges, including hallucinations (misclassifying add-ons as features) and dynamic content limitations (prices requiring user interactions for visibility). Simultaneously, Molino-Peña et al. [68] operationalize LLMs for the detection of abusive clauses in legal contracts (RQ_6: regulatory domain), confronting RQ_3 limitations like semantic ambiguity in dispersed contrac-

tual clauses (contextual fairness of "Excessive Compensation" terms). Both studies corroborate RQ_5's observed trend toward semantic parsing of nested web structures and domain-specific knowledge grounding.

These case studies reinforce RQ_5's evolutionary patterns and RQ_4's evaluation frameworks. They exemplify the SLR-identified shift toward hybrid pipelines (Cavero et al.'s Selenium-LLM extraction; Molino-Peña et al.'s ODRL-SPARQL reasoning) to mitigate RQ_3's integration challenges. Their domain focus–SaaS pricing [11] and consumer law [68]–validates RQ_6's expansion into compliance-sensitive industries. Critically, both adopt RQ_4-prescribed evaluation rigor: Cavero et al. employ precision/recall metrics for feature extraction, while Molino-Peña et al. use SPARQL-based validation to quantify detection accuracy. This synergy underscores RQ_5's trend of structured knowledge representation (Pricing4SaaS metamodels, TOSL ontology) to constrain hallucinations–a persistent RQ_3 limitation.

5 Limitations

This study faces three main limitations. First, our corpus included only journals and conferences, potentially missing relevant innovations from preprints and industry reports due to the rapid pace of LLM research. Second, while the AI-assisted quality assessment showed substantial inter-rater agreement ($\kappa = 0.72$), it might still contain subtle biases related to prompt design. Third, focusing solely on English publications risks underrepresenting important region-specific applications of multilingual models in various global contexts.

6 Conclusions and future work

In this paper, some conclusions can be drawn based on the methodology applied to perform SLR using Prisma. The use of LLM has been adequate to optimize the process of analysis of the collected papers and thus apply the quality questions and leave only the most relevant papers according to the subject matter.

Advances in artificial intelligence have led to an increase in the number of publications related to web scraping or web crawling in the last year. In relation to the authors' countries of origin, there has not been a high number of publications in a specific place.

The most commonly used text extraction technique is web scraping, being JSON the most used format for the output of information in structured format. Finally, it can be seen that most of the publications have been done as applied research.

Incorporating large language models (LLMs) into web scraping and text analysis offers notable benefits, such as improved semantic comprehension, support for multiple languages, and the ability to perform specialized detection tasks. Nevertheless, these advantages come with considerable challenges, including high computational demands, potential biases in data, and ethical concerns. Moving forward, research should aim to overcome these obstacles while maximizing the distinct capabilities LLMs provide for extracting and analyzing web-based information.

The analysis of the 91 studies included in this review provides empirical evidence consistent with an ongoing shift–from rule-based methods toward semantic, LLM-driven approaches in web extraction. Quantitative evidence supports this transition: 94.5% of the analyzed works adopted transformer-based LLMs (BERT, GPT, or LLaMA families), while only 5.5% retained traditional DOM-selector or regex approaches. Moreover, 70.3% of the studies emphasized semantic comprehension and contextual adaptation as explicit objectives, compared to structural accuracy metrics dominant before 2021. This semantic orientation is further corroborated by the emergence of LLM-augmented tools such as Scrapeghost [79] and ChatPhishDetector [52], which automate reasoning and code generation rather than fixed HTML parsing. Collectively, these findings demonstrate a measurable and consistent shift toward context-aware, model-driven extraction pipelines–substantiating that the evolution described in this work is not conceptual but empirically grounded.

SLMs represent a promising direction for future web scraping and crawling research due to their efficiency and reduced computational requirements. Their lightweight architecture enables deployment in low-resource or edge environments while maintaining strong reasoning capabilities. Moreover, SLMs can be fine-tuned on domain-specific web data, supporting adaptive and cost-effective extraction pipelines.

Future systems may combine LLM-based reasoning with traditional scraping frameworks to balance intelligence and efficiency. For instance, LLMs could generate or refine scraping scripts, while tools such as Scrapy or BeautifulSoup handle execution and concurrency. Such hybrid architectures would leverage the adaptability of LLMs with the performance and reliability of established frameworks.

The field also requires standardized benchmarks to ensure fair and reproducible assessment of LLM-assisted scraping systems. Future studies should include metrics for robustness to noisy HTML, adaptability to dynamic layouts, and cost–efficiency trade-offs. Shared datasets and protocols would promote rigorous comparison and accelerate methodological progress.

Supplementary information: The laboratory package of this research is available in Online Resource at the Zenodo repository [56] and includes: 1) the.xlsx file of the dataset used, 2) a.pdf file with the prompt used and tables included as supplementary material.

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Data availability The laboratory package of this research is available in Online Resource at the Zenodo repository [56].

Declarations

Conflict of interest The authors declare no conflict of interest.

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